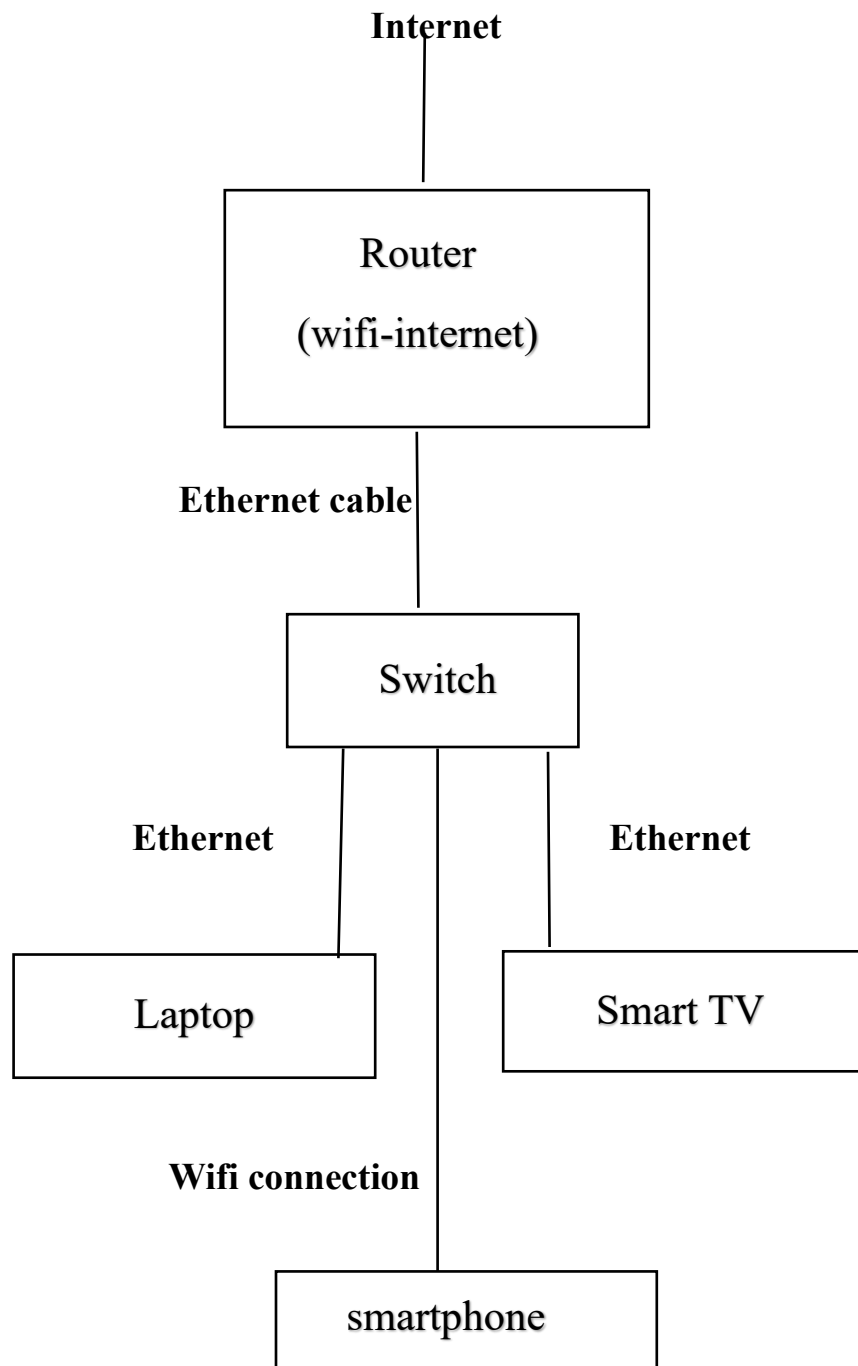


# NETWORK DEVICES

1. Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.



**2. Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).**

| Device            | Use                                                        | Real-World Example                                                                          |
|-------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Router            | Connects devices to the Internet.                          | Used at home to watch YouTube and use Instagram over Wi-Fi.                                 |
| Switch            | Connects computers in the same network (LAN).              | Used in a computer lab for LAN gaming and file sharing.                                     |
| Hub               | Connects multiple computers and sends data to all devices. | Used in small or old office networks for basic file sharing.                                |
| Access Point (AP) | Provides or extends Wi-Fi coverage.                        | Used in colleges and malls so students can use WhatsApp, YouTube, and Google Meet on Wi-Fi. |

**3. Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).**

Scenario: Ordering Food on Zomato

When I open Zomato on my phone and order food, my phone first connects to the Access Point (AP) or the Wi-Fi feature of the router. The router sends my request to the Internet through my Internet Service Provider (ISP). If a switch is used, it connects the router with other wired devices in the network. The request reaches the Zomato server, and the restaurant details are sent back to my phone through the same network path.

**4. Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup.<br><br><em><strong>Hint:</strong> Think about devices like switches, powerline adapters, or WiFi extenders.</em>**

If my gaming PC only has an **Ethernet port** and is far from the Wi-Fi router, I can use these two devices:

### **1. Wi-Fi Extender (Range Extender)**

- A **Wi-Fi extender** receives the Wi-Fi signal from the router and extends its range.
- I can connect my gaming PC to the extender using an **Ethernet cable**.
- This allows the PC to access the Internet even if it is far from the router.

### **2. Powerline Adapter**

- A **powerline adapter** uses the home's electrical wiring to carry the Internet signal.
- One adapter is connected to the router with an Ethernet cable, and the second adapter is plugged in near the gaming PC.
- The PC is then connected to the second adapter using an Ethernet cable.